Practice Questions for 07L2-Vision

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1 Questions

1. According to the lecture, what is the "semantic gap" in the context of image recognition?

   A. The difference in processing speed between humans and computers.

   B. The difference between how humans understand an image and how a computer represents it numerically.

   C. The difference in the number of pixels between high-resolution and low-resolution images.

   D. The difference between the RGB color channels in an image.

   E. The difference between the number of images a human can process versus a computer.

2. In the context of image classification, what is the purpose of finding "invariant features"?

   A. To find features that are unique to each image.

   B. To find features that change with different lighting conditions.

   C. To find features that remain consistent despite variations in the image, such as changes in object position, size, or orientation.

   D. To find features that are only present in high-resolution images.

   E. To find features that are only present in images with a specific color palette.

3. According to the lecture, what is the main idea behind the "Bag of Features" approach for image representation?

   A. To represent an image as a grid of pixels.

   B. To represent an image as a collection of its constituent features, disregarding their spatial relationships.

   C. To represent an image as a sequence of color values.

   D. To represent an image as a set of geometric shapes.

   E. To represent an image as a set of edges and corners.

2 Answers

1. According to the lecture, what is the "semantic gap" in the context of image recognition?

   A. Answer A is incorrect because the semantic gap is not about processing speed.

   B. Answer B is correct because the semantic gap refers to the difference between the numerical representation of an image by a computer and the human understanding of the image's content.

   C. Answer C is incorrect because the semantic gap is not about image resolution.

   D. Answer D is incorrect because the semantic gap is not about the RGB color channels.

   E. Answer E is incorrect because the semantic gap is not about the number of images processed.

2. In the context of image classification, what is the purpose of finding "invariant features"?

   A. Answer A is incorrect because invariant features are not unique to each image, but rather consistent across variations of the same object.

   B. Answer B is incorrect because invariant features should not change with lighting conditions.

   C. Answer C is correct because invariant features are those that remain consistent despite variations in the image.

   D. Answer D is incorrect because invariant features are not specific to high-resolution images.

   E. Answer E is incorrect because invariant features are not specific to a particular color palette.

3. According to the lecture, what is the main idea behind the "Bag of Features" approach for image representation?

   A. Answer A is incorrect because the Bag of Features approach does not focus on the spatial arrangement of pixels.

   B. Answer B is correct because the Bag of Features approach treats an image as a collection of features, disregarding their spatial arrangement.

   C. Answer C is incorrect because the Bag of Features approach does not focus on color values.

   D. Answer D is incorrect because the Bag of Features approach does not focus on geometric shapes.

   E. Answer E is incorrect because the Bag of Features approach does not focus on edges and corners.